

example,

测量.

metricon

传感器如 digital twin.

reality capture. → 精度.

Unreal engine. — out date.

VR

internet things.

lightning maps, real time,

air pollution index

cheap

5G — speed
— 带宽
— deployed.

Water quality — satellite sensors

internet

monitor animal $\rightarrow$ no need hard drive.
SG $\rightarrow$ real time.

Start home.

Week 1 group

VR Systems Benchmark

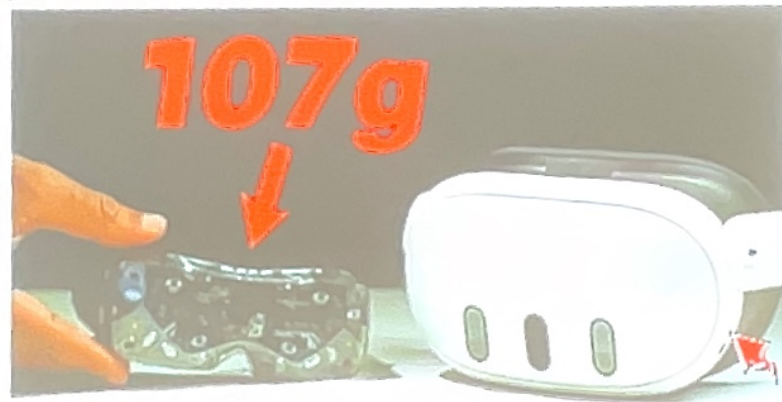
The VR systems are measured with the following categories:

1. Interactivity
2. Usability
3. Immersion

(Dari et al., 2024; Moyle et al., 2018)

VR System Categories & Tech Capabilities

Head-mounted Display Headsets(HMDs) such as Meta's Oculus series or HTC Vive, HMDs are usually high immersion and presence (Markowitz et al., 2018).



VR System Categories & Tech Capabilities

Screen-based (Kinect or Nintendo Wii), it's have lower costs comparing to the other two systems (Moyle et al., 2018).



VR System Categories & Tech Capabilities

Mobile VR, they are very accessible but lower fidelity. It's basically a smartphone strapped to your head (Dari et al., 2024).



Example 1 (Effectiveness of VR in Environmental Education)

Knowledge Gains

The VR group's post-test scores were significantly higher (89%) compared to the control group (76%)

Engagement

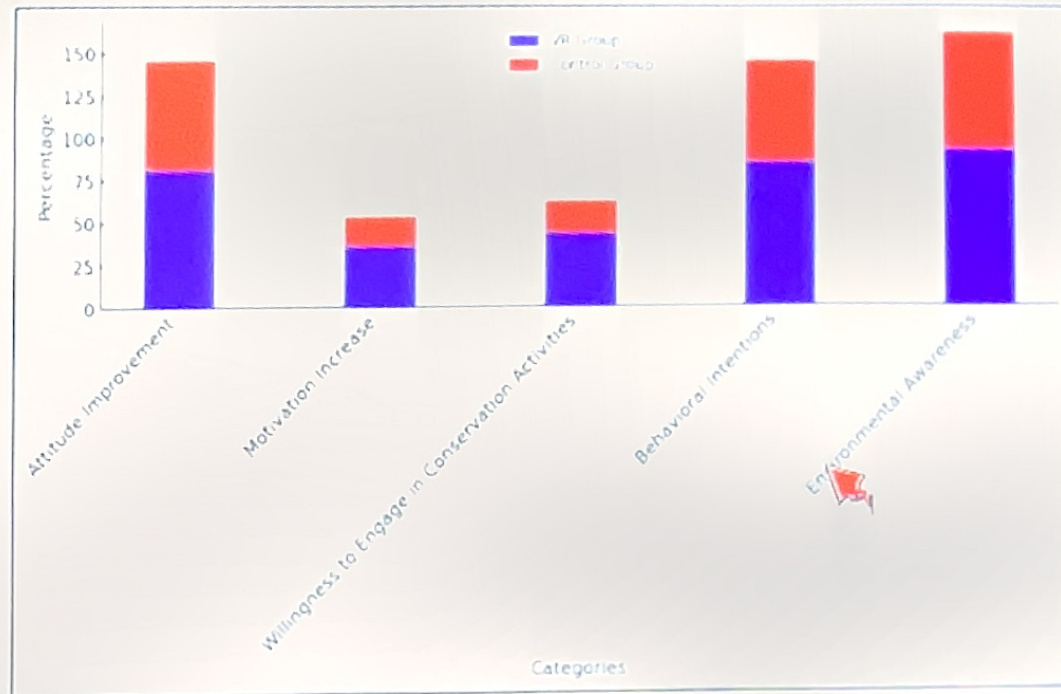
VR students exhibited higher engagement levels (90.2% vs 65.8%)

Attitude

Enhanced positive attitudes and increased willingness to engage in conservation activities were observed in the VR group

Retention

The VR group retained information better over time (85% retention vs 70%)



(Dari et al., 2024)

Example 2-1 Birmingham Zoo & Immotion



Virtual Reality theaters in zoos & aquariums with immersive experiences

INDOOR



4 - 48+ Seats

Full, 360° panoramic views of theaters

(www.immotion.co, 2025)

Example 2-2 Wild Immersion

no reading
no interaction

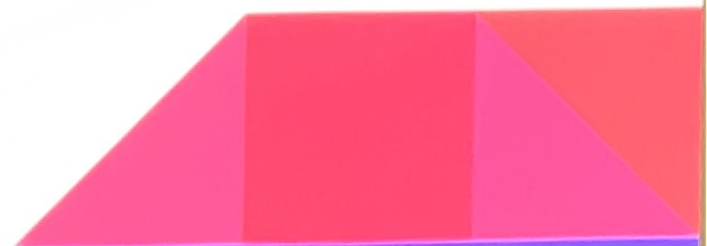
Wild Immersion VR solution in traffic hubs, raising conservation awareness as well enhancing waiting experience



(wildimmersion.io, 2025)

Implementation Determinants & Barriers

1. Determinants: technological readiness, engagement and interactivity, knowledge acquisition (Dari et al., 2024)
2. Barriers: cost and accessibility, technological limitations and training, content development, adaptation to different learning environments (Dari et al., 2024)



Conclusion & Future Direction

1. VR enhances environmental knowledge, engagement, and attitudes (Dari et al., 2024; Markowitz et al., 2018)
2. Scale challenges: cost, training, accessibility (Dari et al., 2024)
3. Need longitudinal behavior-change studies & low-cost VR (Markowitz et al., 2018)
4. VR's potential as a core global environmental education tool

interaction

quiz

Summary

- VR creates immersive, interactive experiences that boost environmental knowledge, engagement, and pro-environmental attitudes more than traditional methods (Dari et al., 2024; Markowitz et al., 2018).
- Real-world examples (Wild Immersion, Birmingham Zoo's Wild Explorer) illustrate VR's potential to raise conservation awareness and support.
- Key barriers include high hardware costs, required technical training, accessibility gaps, and fading novelty effects.
- Future priorities: develop low-cost VR solutions, conduct longitudinal studies on lasting behavior change, and implement equitable integration strategies for diverse educational settings.

title

finding → make source

over view of research before detail